

1	c	eg $12 \times \frac{4x-2}{3} - 12 \times \frac{5-3x}{4} = 12 \times 6$ or eg $4(4x-2) - 3(5-3x) = 12 \times 6$ or eg $\frac{4(4x-2)}{12} - \frac{3(5-3x)}{12} (=6)$ or eg $\frac{4(4x-2) - 3(5-3x)}{12} (=6)$ oe			M1 for clear intention to multiply all terms by 12 or a multiple of 12 or to express LHS as two fractions over 12 or a multiple of 12 or as a single fraction with a denominator of 12 or a multiple of 12 (if expanded numerator, allow one sign error)
		eg $16x - 8 - 15 + 9x = 6 \times 12$			M1 expanding brackets and multiplying both sides by denominator with no more than one sign error
		eg $16x + 9x = 72 + 8 + 15$			M1 for correct rearrangement of a correct equation with terms in x isolated
			3.8	4	A1 oe, award full marks for a correct answer if at least M1 scored

2	(d)	E.g. $6x - 15$ or $12x - 30$ oe		4	M1 for expansion of a correct bracket
		$2 \times 3(2x - 5) = 9 - x$ oe or $2('6x - 15') = 9 - x$ oe or $3(2x - 5) = \frac{9}{2} - \frac{x}{2}$ oe			M1 for removal of fraction or separating fraction (RHS) in an equation
		$12x + x = 9 + 30$ oe or $6x + \frac{x}{2} = \frac{9}{2} + 15$ oe			M1 ft (dep on 4 terms) for terms in x on one side of equation; number terms on the other
			3		A1 dep on at least M2 awarded

3	(c)	$w^2 \times w^n = w^{10}$ or $w^5 \times w^n = w^{13}$ or $w^5 \times w^{n-3} = w^{10}$ or $\frac{w^{5+n}}{w^3} = w^{10}$ oe or $5 + n - 3 = 10$ or $2 + n = 10$ or $5 + n = 13$		2	M1 A correct first stage simplifying at least one index in a correct equation or a clearly correct subsequent stage showing correct use of a rule of indices eg $w^5 \times w^n = w^{30}$ and $w^n = w^{30-5}$ or a correct equation using indices only
		<i>Working not required, so correct answer scores full marks (unless from obvious incorrect working)</i>	8		A1 accept w^8 (trial and error gains full marks if correct and no marks if incorrect unless a rule of indices is clearly shown)

4	c	$(4^{k+3}) = (2^2)^{k+3}$ oe or $(16 =) 2^4$ $(16 =) 4^2$ or $(2^k =) \left(4^{\frac{1}{2}}\right)^k$ oe $(4^{k+3} =) \left(16^{\frac{1}{4}}\right)^{k+3}$ oe or $(2^k =) \left(16^{\frac{1}{4}}\right)^k$ oe		4	M1 for $(2^2)^{k+3}$ oe or 2^4 or 4^2 or $\left(4^{\frac{1}{2}}\right)^k$ oe or $\left(16^{\frac{1}{4}}\right)^{k+3}$ oe or $\left(16^{\frac{1}{4}}\right)^k$ oe
		$(4^{k+3} =) (2^2)^{k+3}$ oe and $(16 =) 2^4$ $(16 =) 4^2$ and $(2^k =) \left(4^{\frac{1}{2}}\right)^k$ oe $(4^{k+3} =) \left(16^{\frac{1}{4}}\right)^{k+3}$ oe and $(2^k =) \left(16^{\frac{1}{4}}\right)^k$ oe			M1 for $(2^2)^{k+3}$ oe and 2^4 or 4^2 and $\left(4^{\frac{1}{2}}\right)^k$ oe or $\left(16^{\frac{1}{4}}\right)^{k+3}$ oe and $\left(16^{\frac{1}{4}}\right)^k$ oe
		E.g. $2k + 6 = 4 + k$ or $k + 3 = 2 + \frac{1}{2}k$ or $\frac{1}{2}(k + 3) = 1 + \frac{1}{4}k$			M1 for a correct linear equation in k
			-2		A1 dep on at least M2

5	(b)	eg $y^5 \times y^n = y^{19}$ or $y^{-1} \times y^n = y^{13}$ or $5 + n - 6 = 13$	14	2	M1 Use of 1 rule of indices or a correct linear equation in n
					A1 Accept y^{14}